

CARE-PROGRESS India

Website Build Specification v1.0

Survey Website for Packet-Collected Outpatient Care-Loop Progression Data

Core build objective

Build a mobile-first, de-identified survey website that collects specific recent outpatient care-loop audits from doctors and clinic staff. Each submission must produce analyzable care-loop rows with clinical urgency, event timing offsets, workflow role path, handoff count, tracking method, resolution status, evidence source, and confidence.

Spec dimension	Required decision
Primary object	Care loop, not patient, not doctor opinion, not generic workflow impression.
Required respondent task	Complete context module and audit 6 recent care loops: ideally 3 abnormal diagnostic result loops and 3 referral/follow-up loops.
Target completion time	13-19 minutes; hard cap 25 minutes.
Collection window	Two-week pilot, with one-week field push possible once the website is live.
Privacy posture	No patient identifiers, no raw clinical documents, no exact identifiable dates when avoidable; use loop codes and relative day offsets.
Core outputs	Clean CSV exports for respondents, care loops, event timelines, workflow paths, resolution outcomes, evidence quality, and site operations.

Source basis: this website specification operationalizes the CARE-PROGRESS India field protocol uploaded as the master research instrument. The field protocol frames CARE-PROGRESS as a packet-collected dataset for abnormal diagnostic results, referrals, follow-up loops, and risk-weighted delay.

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1. Executive Summary and Build Philosophy

CARE-PROGRESS India is not a normal survey site. It is a structured data-collection instrument for time-sensitive outpatient care progression. The website must force each useful record to be anchored to one specific recent care loop, not to a general impression of clinic workflow.

Final operating principle

Never ask what usually happens. Ask what happened in one specific recent care loop. Capture urgency, timeline, workflow path, resolution, evidence, and confidence. Then repeat.

The source field protocol defines the scientific object as care progression under clinical time-sensitivity and uncertainty. The website must preserve that object by collecting loop-level event timelines, clinical urgency, handoffs, role transitions, tracking systems, closure status, and evidence quality. This enables regression, survival analysis, temporal sequence modeling, workflow graph learning, stall-stage prediction, and observational counterfactual analysis.

Weak implementation	Correct implementation
Doctor opinion survey	Specific loop audit using records where possible
One binary follow-up label	Event sequence with review/contact/action/closure offsets
Exact patient-level dates and details	Relative day offsets and de-identified loop codes
Generic delay measurement	Risk-weighted delay under clinical urgency
Single giant form blob	Structured data rows with exportable schema

2. Product Scope: What the Website Must Collect

The MVP website collects one respondent/session record and at least six care-loop rows. The repeated care-loop module is the core of the product. All other modules exist to contextualize or validate those loop rows.

Scope item	MVP requirement	Reason
Consent and de-identification	Mandatory gate before form access	Prevents identifiable data entry and clarifies research-only status.
Respondent/site context	One short module per respondent	Provides predictors such as specialty, setting, tracking method, and digital maturity.
Care-loop audit	Six required loops, optional up to ten	Creates analyzable rows for delay, survival, sequence, and graph tasks.
Operational structure	One short module after loop entry	Captures named ownership, unresolved-loop review cadence, and future data export feasibility.
Verification willingness	Optional question at the end	Creates a future validation pathway for a gold subset.
Export	Downloadable CSVs or database export	Makes the website useful for analysis immediately.

Non-goals for v1

Do not build login systems, doctor dashboards, AI models, clinical decision support, hospital account management, fancy analytics, payment flows, file upload, or messaging. The v1 product is a disciplined survey-to-CSV instrument.

3. User Personas and Permissions

Persona	Primary use	Design implication
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Physician	Supplies clinical urgency, severity, care consequence proxies, and clinical review/action interpretation.	Language must be clinically precise and fast. Avoid administrative clutter.
Nurse / medical assistant	Knows patient contact, follow-up logistics, action completion, and care-loop status.	Allow non-physician respondents. Do not assume only doctors can complete loops.
Clinic coordinator / receptionist	Often knows scheduling, contact attempts, referrals, and closure status.	Use role categories, not clinician-only terminology.
Lab/referral coordinator	Knows result availability and referral completion pathways.	Loop family and role fields must include lab/referral staff.
Site champion/admin	Can provide site operations and future export feasibility.	Admin responses are valuable for siteOperations but not sufficient for loop-level labels alone.
Research team/admin	Monitors response counts, exports CSV, grades rows, and checks privacy.	Needs minimal admin console or backend export access.

No end user should see patient names, exact patient identifiers, or any file upload prompt. Named staff members are also unnecessary. Role-level graphs are sufficient.

4. End-to-End User Flow

17. Landing page explains the study, time burden, care-loop definition, and privacy boundary.
18. Consent gate requires de-identification confirmation and research-only acknowledgment.
19. Respondent/site context collects role, specialty, setting, record system, tracking methods, communication channel, and digital maturity.
20. Loop dashboard shows 6 required loops, recommends 3 abnormal diagnostic result loops and 3 referral/follow-up loops, and allows optional extra loops.
21. Repeated loop audit collects loop identity, clinical time-sensitivity, event timeline, workflow role path, workflow structure, resolution/consequence proxies, and evidence quality.
22. Operational structure module asks about named owners, unresolved-loop review cadence, future data export feasibility, and approval path.
23. Review page checks missing required fields and impossible timelines before submit.
24. Thank-you page gives no clinical advice and optionally invites future participation.

Progress indicator

Use a persistent progress indicator: Intro -> Consent -> Context -> Loop 1 -> Loop 2 -> Loop 3 -> Loop 4 -> Loop 5 -> Loop 6 -> Operations -> Submit. Do not show a frightening 80-question progress bar.

5. Screen-by-Screen Specification

Screen 1 - Landing / Study Explanation

Element	Specification
Title	CARE-PROGRESS India: Outpatient Care-Loop Progression Audit
Subtitle	A de-identified survey website for abnormal diagnostic result, referral, and follow-up loop progression.
Estimated time	13-19 minutes. Six loop audits required.
Primary CTA	Start de-identified audit
Secondary note	Use records where possible. Do not enter patient-identifying information.

Landing copy

CARE-PROGRESS India studies how time-sensitive outpatient care loops move from event creation to review, patient contact, action, escalation, and closure.

You will be asked to audit 6 recent care loops from the last 30-90 days. A care loop may be an abnormal diagnostic result, referral, follow-up request, repeat test order, post-procedure follow-up, or patient-contact task.

Use records where possible. Do not enter patient names, phone numbers, addresses, Aadhaar numbers, medical record numbers, screenshots, prescriptions, lab-report photos, or other identifying information.

Screen 2 - Consent and De-identification Gate

Field key	Prompt	Input type	Required	Validation/behavior
confirm_no_identifiers	I confirm that I will not enter patient names, phone numbers, addresses, Aadhaar numbers, medical record numbers, prescription images, lab-report photos, WhatsApp screenshots, raw clinical notes, or other identifiable information.	Checkbox	Yes	Must be checked to continue.
confirm_research_only	I understand this is a de-identified research audit of outpatient care-loop progression, not a doctor performance evaluation and not a clinical decision-making tool.	Checkbox	Yes	Must be checked to continue.
record_use_basis	Are you using clinic records where possible?	Radio	Yes	Options: Mostly records / Mixed records and memory / Mostly memory.

Screen 3 - Respondent and Site Context

Field key	Prompt	Options	Required
respondent_role	Respondent role	Physician / Nurse / Medical assistant / Clinic coordinator / Receptionist / Lab staff / Admin /	Yes

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		Other	
specialty	Specialty or service area	General medicine / Cardiology / Endocrinology / Oncology / OB-GYN / Pediatrics / Surgery / Orthopedics / Neurology / Multi-specialty / Other	Yes
practice_setting	Practice setting	Solo private clinic / Group private clinic / Private hospital OPD / Government hospital OPD / Charitable clinic / Telemedicine / Other	Yes
opd_volume	Average outpatient volume per day	<20 / 20-39 / 40-59 / 60-79 / >=80 / Unknown	Yes
support_staff	Support staff usually available	Receptionist / Nurse / MA / Coordinator / Lab coordinator / Pharmacist / None / Other	Yes
record_system	Main record system	Paper / EHR / Clinic software / WhatsApp/photos / Mixed / Other	Yes
lab_tracking	Lab/result tracking method	None / Physician memory / Staff memory / Paper log / Spreadsheet / Clinic software / EHR task / WhatsApp / Mixed / Unknown	Yes
referral_tracking	Referral tracking method	Same options as lab tracking	Yes
followup_tracking	Follow-up tracking method	Same options as lab tracking	Yes
communication_channel	Main patient communication channel	Phone / WhatsApp / SMS / In-person / Portal/app / Email / Other	Yes
digital_maturity	Digital maturity score	1 mostly paper / 2 paper + phone / 3 spreadsheet or WhatsApp tracking / 4 clinic software / 5 integrated EHR or task system	Yes

Screen 4 - Loop Audit Dashboard

- Display requirement: 6 required loops. Recommended mix: 3 abnormal diagnostic result loops and 3 referral/follow-up loops.
- Each loop card shows status: Not started / In progress / Complete / Needs correction.
- Allow optional Add another loop after six complete loops, up to ten total loops.
- Show a short reminder: Use most recent consecutive eligible loops from the last 30-90 days. Do not choose only memorable, severe, successful, or failed cases.

Dashboard component	Behavior
Loop cards	Six cards visible, labeled Loop 1 through Loop 6. Each opens the repeated care-loop form.
Completion check	Loop card becomes complete only when required fields pass validation.
Mix hint	If all first 3 loops are same family, show non-blocking reminder to include both loop families when possible.
Autosave indicator	Show Saved or Draft saved after each loop.

Screen 5 - Repeated Care-Loop Form

This screen is specified in detail in Section 6. It should be implemented as a reusable component that can be rendered for Loop 1 through Loop 10. Every loop must store its own values and validation state.

Screen 6 - Operational Structure

Field key	Prompt	Options	Required
abnormal_result_owner_exists	Does the clinic have a named	Yes / No / Sometimes / Unknown	Yes

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	person responsible for abnormal result follow-up?		
referral_owner_exists	Does the clinic have a named person responsible for referral completion?	Yes / No / Sometimes / Unknown	Yes
unresolved_review_cadence	Are unresolved loops reviewed at the end of the day/week?	Daily / Weekly / Occasionally / No / Unknown	Yes
future_export_available	Can the clinic export de-identified logs in the future?	Appointments / labs / referrals / follow-ups / contact logs / none / unknown	Yes
future_approval_path	Who would approve future data sharing?	Doctor / clinic owner / hospital admin / ethics committee / IT department / unknown	Yes

Screen 7 - Review and Submit

- Summarize context completeness and number of complete loops.
- Flag required missing fields before submit.
- Flag impossible timelines and ask respondent to correct or mark unknown.
- Show privacy reminder one final time before submit.
- Submit creates respondent_id, submission_id, timestamps, and loop_id values for each loop.

Screen 8 - Thank You

Element	Specification
Message	Thank you for contributing to CARE-PROGRESS India. Your submission helps measure how outpatient care loops progress, stall, and close across clinical workflows.
No clinical advice	Do not display any patient-specific recommendations or model output.
Optional future contact	If future follow-up contact is desired, collect it in a separate form not linked to loop rows, or omit entirely in the pilot.

6. Care-Loop Audit Module: Field Details

The care-loop audit module is the central reusable component. It must be concise enough to complete in 90-150 seconds per loop while still preserving temporal, clinical, workflow, and evidence structure.

Loop Identity

Field key	Prompt	Response options / entry	Required
loop_code	Loop code	Text default L001, L002, etc. Do not use patient identifiers.	Yes
loop_family	Loop family	Abnormal diagnostic result / Referral or follow-up	Yes
loop_type	Specific loop type	Abnormal lab / Abnormal imaging / Abnormal pathology / Specialist referral / Imaging referral / Procedure referral / Follow-up appointment / Repeat test / Post-procedure follow-up / Other	Yes
loop_specialty	Specialty involved	General medicine / Cardiology / Endocrinology / Oncology / OB-GYN / Pediatrics / Surgery / Orthopedics / Neurology / Other	Yes
test_category	For abnormal result loops: test category	CBC / HbA1c / renal function / liver function / thyroid / lipid / X-ray / ultrasound / CT/MRI / mammogram / pathology / other / N/A	Conditional
referral_category	For referral/follow-up loops: category	Cardiology / oncology / endocrinology / imaging / surgery / OB-GYN / pediatrics / post-procedure / chronic follow-up / other / N/A	Conditional

Clinical Time-Sensitivity

Field key	Prompt	Response options / entry	Required
urgency	Clinical urgency	Routine / Semi-urgent / Urgent / Critical / Unclear	Yes
severity	Clinical severity	Mild / Moderate / Severe / Critical / Not applicable / Unclear	Yes
recommended_window	Recommended closure window	Same day / 1-3 days / 4-7 days / 8-14 days / 15-30 days / >30 days / Unclear	Yes
delay_affect_care	Could delay plausibly affect care?	No / Possibly / Yes / Unclear	Yes

Event Timeline

Field key	Prompt	Response options / entry	Required
review_status	Clinician review status	Reviewed / Not reviewed / Not needed / Unknown	Yes
review_day_offset	Clinician review day offset	Numeric days after Day 0 if known	Conditional
contact_status	Patient contact status	Contacted / Not contacted / Not needed / Unknown	Yes
contact_day_offset	Patient contact day offset	Numeric days after Day 0 if known	Conditional
action_status	Documented action status	Action documented / No action needed / Action not documented / Unknown	Yes
action_day_offset	Documented action day offset	Numeric days after Day 0 if known	Conditional
followup_scheduled_status	Follow-up scheduled status	Scheduled / Not scheduled / Not needed / Unknown	Yes

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followup_scheduled_day_offset	Follow-up scheduled day offset	Numeric days after Day 0 if known	Conditional
closure_status	Loop closure status	Closed / Still open / Lost to follow-up / Escalated / Closed as no action needed / Unknown	Yes
closure_day_offset	Loop closure day offset	Numeric days after Day 0 if closed	Conditional
audit_age_days_if_open	If still open: audit age in days	Numeric days old at audit if known	Conditional

Workflow Path / Graph Fields

Field key	Prompt	Response options / entry	Required
first_receiver_role	Who first received or noticed the loop?	System alert / Physician / Nurse / Receptionist / Coordinator / Lab staff / Patient / Unknown	Yes
reviewer_role	Who reviewed it clinically?	Physician / Nurse / Specialist / Not reviewed / Unknown	Yes
contact_role	Who contacted the patient?	Physician / Nurse / Receptionist / Coordinator / Patient not contacted / Not needed / Unknown	Yes
owner_role	Who primarily owned closure?	Physician / Nurse / Coordinator / Receptionist / Patient / No clear owner / Unknown / Other	Yes
closer_role	Who closed the loop?	Physician / Nurse / Coordinator / Receptionist / System / Still open / Unknown	Yes
handoff_count	Number of handoffs	0 / 1 / 2 / 3 / 4+ / Unknown	Yes

Workflow Structure

Field key	Prompt	Response options / entry	Required
tracking_method	Tracking method for this loop	No formal tracking / Physician memory / Staff memory / Paper log / Spreadsheet / Clinic software / EHR task / WhatsApp / Mixed / Unknown	Yes
support_staff_involved	Support staff involved?	Yes / No / Unclear	Yes
loop_communication_channel	Main communication channel	Phone / WhatsApp / SMS / In-person / Portal/app / Other / Unknown	Yes
clinic_load	Approximate outpatient load that day	<20 / 20-39 / 40-59 / 60-79 / >=80 / Unknown	Recommended

Resolution and Clinical Consequence Proxy

Field key	Prompt	Response options / entry	Required
current_status	Current status	Resolved / Still open / Lost to follow-up / Escalated / Closed as no action needed / Unknown	Yes
patient_contacted	Was the patient successfully contacted?	Yes / No / Not needed / Unknown	Yes
followup_completed	Was follow-up completed?	Yes / No / Not needed / Unknown	Yes
escalation_required	Was urgent escalation required?	Yes / No / Unclear	Yes
emergency_referral_or_admission	Was emergency referral/admission documented?	Yes / No / Unclear	Yes
care_plan_changed	Was the care plan changed because of this loop?	Yes / No / Unclear	Yes

Evidence Quality

Field key	Prompt	Response options / entry	Required
evidence_source	Evidence source	EHR / Paper chart / Lab log / Referral register / Appointment register / Staff call log / Physician memory / Staff memory / Other	Yes
confidence	Confidence	Low / Medium / High	Yes
selection_method	Selection method	Most recent consecutive loops / Randomly selected loops / Convenience sample / High-risk examples only / Unknown	Yes
verification_willingness	May a de-identified subset be verified later by an abstractor?	Yes / No / Maybe	Optional

Relative day offset guidance

The index event is Day 0. If an event happened the same day, enter 0. If it happened three days later, enter 3. If the event did not happen, was not needed, or is unknown, use the status field rather than inventing a number.

7. Conditional Logic and Validation Rules

7.1 Conditional display rules

Condition	Show / require
loop_family = Abnormal diagnostic result	Show test_category. Hide or default referral_category to N/A.
loop_family = Referral or follow-up	Show referral_category. Hide or default test_category to N/A.
review_status = Reviewed	Require review_day_offset unless respondent selects Unknown offset.
contact_status = Contacted	Require contact_day_offset unless respondent selects Unknown offset.
action_status = Action documented	Require action_day_offset unless respondent selects Unknown offset.
followup_scheduled_status = Scheduled	Require followup_scheduled_day_offset unless unknown.
closure_status in Closed, Resolved, Closed as no action needed	Require closure_day_offset unless unknown.
closure_status = Still open	Require audit_age_days_if_open when known; set censored_status = 1.

7.2 Blocking validation rules

Rule	Error message / behavior
No consent checkbox checked	Block progress: Please confirm de-identification and research-only use before continuing.
Loop missing family/type/urgency/window/status/evidence/confidence/selectio n_method	Mark loop incomplete and block final submission.
Day offset contains text other than allowed unknown marker	Require numeric day offset or mark the event status as Unknown/Not needed.
Day offset < 0 while event status indicates event occurred	Show error. Use Unknown status instead of negative day values.
closure_day_offset < review_day_offset when both known and closure implies review/action occurred	Warn and block unless respondent changes one status to Unknown or corrects value.
loop_code appears to contain obvious patient identifier patterns	Warn: Do not use patient identifiers. Ask respondent to replace with L001-style code.

7.3 Non-blocking warnings

- If all six loops are one family, warn that a mix is preferred but allow submission.
- If evidence source is memory-only, allow submission but mark lower evidence grade.
- If selection method is High-risk examples only, allow submission but flag row as not prevalence-estimable.
- If many offsets are unknown, allow submission but flag Grade C or D during quality grading.

8. Data Model and Database Schema

The fastest MVP can store a single wide care_loops table, but the logical design should follow relational boundaries so later graph and sequence modeling remain feasible. Each respondent has one respondent context record and multiple care_loop records.

Logical table	Purpose	Key fields
respondents	One row per survey session/respondent.	respondent_id, submission_id, role, specialty, practice_setting, record_system, tracking methods, digital_maturity, consent flags
care_loops	One row per audited care loop.	loop_id, respondent_id, loop_family, loop_type, urgency, severity, recommended_window, selection_method
event_timeline	Timeline statuses and offsets.	loop_id, review_status/day, contact_status/day, action_status/day, followup_status/day, closure_status/day, audit_age_days_if_open
workflow_path	Role path and handoffs.	loop_id, first_receiver_role, reviewer_role, contact_role, owner_role, closer_role, handoff_count
resolution_outcomes	Closure and consequence proxies.	loop_id, current_status, patient_contacted, followup_completed, escalation_required, emergency_referral_or_admission, care_plan_changed
evidence_quality	Evidence and confidence metadata.	loop_id, evidence_source, confidence, record_use_basis, verification_willingness
site_operations	One row per respondent/site operations context.	respondent_id, named owners, unresolved review cadence, future export availability, approval path

8.1 Suggested SQL schema for MVP

```
-- respondents
respondent_id UUID PRIMARY KEY
submission_id UUID NOT NULL
created_at TIMESTAMP NOT NULL
record_use_basis TEXT NOT NULL
respondent_role TEXT NOT NULL
specialty TEXT NOT NULL
practice_setting TEXT NOT NULL
opd_volume TEXT NOT NULL
support_staff TEXT[]
record_system TEXT NOT NULL
lab_tracking TEXT NOT NULL
referral_tracking TEXT NOT NULL
followup_tracking TEXT NOT NULL
communication_channel TEXT NOT NULL
digital_maturity INTEGER NOT NULL

-- care_loops
loop_id UUID PRIMARY KEY
respondent_id UUID REFERENCES respondents(respondent_id)
loop_number INTEGER NOT NULL
loop_code TEXT NOT NULL
loop_family TEXT NOT NULL
loop_type TEXT NOT NULL
loop_specialty TEXT NOT NULL
test_category TEXT
referral_category TEXT
```

urgency TEXT NOT NULL
severity TEXT NOT NULL
recommended_window TEXT NOT NULL
delay_affect_care TEXT NOT NULL
review_status TEXT NOT NULL
review_day_offset INTEGER
contact_status TEXT NOT NULL
contact_day_offset INTEGER
action_status TEXT NOT NULL
action_day_offset INTEGER
followup_scheduled_status TEXT NOT NULL
followup_scheduled_day_offset INTEGER
closure_status TEXT NOT NULL
closure_day_offset INTEGER
audit_age_days_if_open INTEGER
first_receiver_role TEXT NOT NULL
reviewer_role TEXT NOT NULL
contact_role TEXT NOT NULL
owner_role TEXT NOT NULL
closer_role TEXT NOT NULL
handoff_count TEXT NOT NULL
tracking_method TEXT NOT NULL
support_staff_involved TEXT NOT NULL
loop_communication_channel TEXT NOT NULL
clinic_load TEXT
current_status TEXT NOT NULL
patient_contacted TEXT NOT NULL
followup_completed TEXT NOT NULL
escalation_required TEXT NOT NULL
emergency_referral_or_admission TEXT NOT NULL
care_plan_changed TEXT NOT NULL
evidence_source TEXT NOT NULL
confidence TEXT NOT NULL
selection_method TEXT NOT NULL
verification_willingness TEXT

9. CSV Export Specifications

At minimum, export respondents.csv and care_loops.csv. If engineering time allows, export logical split tables as separate files. The wide care_loops.csv must contain enough columns to reconstruct the split tables.

CSV file	Required columns
respondents.csv	respondent_id, submission_id, created_at, record_use_basis, respondent_role, specialty, practice_setting, opd_volume, support_staff, record_system, lab_tracking, referral_tracking, followup_tracking, communication_channel, digital_maturity
care_loops.csv	loop_id, respondent_id, loop_number, loop_code, loop_family, loop_type, loop_specialty, test_category, referral_category, urgency, severity, recommended_window, delay_affect_care, all timeline/status/path/outcome/evidence fields
event_timeline.csv	loop_id, review_status, review_day_offset, contact_status, contact_day_offset, action_status, action_day_offset, followup_scheduled_status, followup_scheduled_day_offset, closure_status, closure_day_offset, audit_age_days_if_open
workflow_path.csv	loop_id, first_receiver_role, reviewer_role, contact_role, owner_role, closer_role, handoff_count, tracking_method, support_staff_involved, loop_communication_channel, clinic_load
resolution_outcomes.csv	loop_id, current_status, patient_contacted, followup_completed, escalation_required, emergency_referral_or_admission, care_plan_changed
evidence_quality.csv	loop_id, evidence_source, confidence, selection_method, verification_willingness, record_use_basis
site_operations.csv	respondent_id, abnormal_result_owner_exists, referral_owner_exists, unresolved_review_cadence, future_export_available, future_approval_path

Export invariant

Every care-loop row must include respondent_id and loop_id. No row should require patient identity to be interpretable. No raw free text should be necessary for primary analysis.

10. Derived Variables and Analytics Logic

Derived variable	Definition
days_to_review	review_day_offset if review_status = Reviewed and offset is known.
days_to_contact	contact_day_offset if patient contact was needed and known.
days_to_action	action_day_offset if action was documented and known.
days_to_followup_scheduled	followup_scheduled_day_offset if scheduled and known.
days_to_closure	closure_day_offset if closure_status indicates closed/resolved/closed as no action needed and offset is known.
censored_status	1 if loop is still open at audit or closure time is unknown but event is not known to be closed; 0 if closure time known.
censored_time	audit_age_days_if_open for still-open loops.
recommended_window_days	Convert window category to representative value: same day=0, 1-3=3, 4-7=7, 8-14=14, 15-30=30, >30=60 unless locally specified.
days_overdue	days_to_closure - recommended_window_days, only when closure time is known.
positive_days_overdue	max(days_overdue, 0).
urgency_weight	Routine=1, Semi-urgent=2, Urgent=3, Critical=5, Unclear=missing.
risk_weighted_delay	positive_days_overdue * urgency_weight.
stall_stage	Stage where progression stopped or largest delay accumulated: pre-review, contact, action, scheduling, closure, referral completion, documentation unknown.

10.1 Suggested stall-stage logic

Condition	stall_stage
review_status = Not reviewed and closure_status not closed	pre_review
reviewed but contact needed and contact_status = Not contacted	contact
contacted but action_status = Action not documented and action needed	action
action documented but followup_scheduled_status = Not scheduled when follow-up needed	scheduling
follow-up scheduled but current_status still open/lost/unknown	closure_or_followup_completion
closure_status = Lost to follow-up	lost_to_followup
critical fields unknown	documentation_unknown

11. Privacy, Security, and De-identification

- Do not collect patient names, phone numbers, addresses, Aadhaar numbers, medical record numbers, prescription images, raw notes, WhatsApp screenshots, lab-report photos, or exact identifiable dates tied to rare events.
- Use loop codes such as L001, L002, L003. Never ask respondents to paste patient IDs into loop_code.
- Use relative day offsets instead of exact dates whenever possible.
- Use role categories rather than staff names.
- Use broad specialties and categories rather than rare identifiable clinical details.
- Remove or redact any accidental identifiers before analysis.
- This pilot must not be used for clinical decision support, doctor performance evaluation, or individual clinic ranking.

Security requirement	MVP implementation
Transport security	Deploy over HTTPS only. Vercel/Supabase provide HTTPS by default.
Database access	Restrict database exports to research/admin account. Do not expose public read APIs.
File uploads	Disable entirely.
Free text	Minimize. Add patient-identifier warning above every optional text field.
Audit logs	Record created_at and updated_at for submissions if possible.
Deletion/redaction	Build an admin process to delete rows containing accidental identifiers.

12. Admin Console and Operations

The admin console for v1 can be extremely simple. If engineering time is tight, Supabase table view plus a CSV export button is enough. Do not spend field time building dashboards.

Admin capability	MVP behavior
View totals	Number of respondents, complete respondents, total loop rows, complete loop rows.
Filter rows	By evidence source, confidence, selection method, loop family, urgency, status.
Export CSV	Export respondents.csv and care_loops.csv at minimum.
Privacy scan	Flag free-text fields and loop_code values for manual review.
Delete/redact row	Allow admin to remove rows with identifiers or severe data quality problems.
Quality grade	Optional computed grade A/B/C/D based on evidence, confidence, completeness, and specificity.

13. UX/UI Requirements and Copywriting Rules

- Mobile-first layout. Assume respondents may use phones in clinics.
- Use large tap targets, short pages, and clear progress.
- Prefer radio buttons, dropdowns, and segmented controls. Avoid long text boxes.
- Show examples next to day-offset fields: same day = 0, three days later = 3.
- Use neutral language. Do not imply blame, negligence, or doctor performance scoring.
- Use record-backed language: Use records where possible; if not available, mark memory and confidence honestly.
- Do not use flashy visual effects. The site should feel credible, restrained, and fast.

Bad copy	Better copy
Did your clinic fail to follow up?	What was the current status of this specific care loop?
Was this doctor responsible?	Who primarily owned closure?
How often are referrals delayed?	For this specific referral/follow-up loop, when did closure occur?
Was the patient harmed?	Was urgent escalation, emergency referral/admission, or care-plan change documented?
Upload the report	Select evidence source: lab log, paper chart, EHR, referral register, memory, or other.

14. MVP Implementation Plan

Day	Engineering task	Acceptance marker
Day 1 morning	Create Next.js/React project, Supabase project, table schemas, env config.	App loads, DB connection works.
Day 1 afternoon	Build Landing, Consent, Context, Loop Dashboard, Loop Form components.	A respondent can complete one loop end-to-end.
Day 2 morning	Implement six required loop flow, validation, draft storage, final submit.	Complete submission creates respondent and six loop rows.
Day 2 afternoon	CSV export, admin access, privacy review, mobile test, fake data test.	Researcher can export analyzable CSV; no file uploads or identifier fields exist.
Day 3	Pilot with 3-5 doctors/staff and revise confusing fields.	Each loop takes under 2.5 minutes; day offsets understood.

Build rule

If custom build will consume more than two field days, use a no-code fallback. The scarce resource is collection time, not software elegance.

15. QA Test Plan and Acceptance Criteria

15.1 Functional tests

- User cannot proceed without de-identification confirmation.
- User can complete respondent context in under 3 minutes.
- User can add, edit, and complete six care loops.
- Conditional fields show correctly for abnormal result vs referral/follow-up loops.
- Final submit is blocked when required fields are missing.
- CSV export contains one respondent row and six care-loop rows per complete submission.
- No file upload fields exist anywhere.

15.2 Data validation tests

Test case	Expected behavior
closure_day_offset entered as -5	Block or reject.
review_status = Reviewed but review_day_offset blank	Require offset or allow explicit Unknown offset state.
closure_day_offset earlier than review_day_offset	Warn/block unless values changed to Unknown or corrected.
loop_code = patient phone number pattern	Warn and require replacement.
evidence_source missing	Block final loop completion.
confidence missing	Block final loop completion.
Only 5 loops complete	Block final submission or clearly mark partial if admin enables partial mode.

15.3 Research acceptance criteria

- At least 90% of complete submissions have six loop rows.
- At least 70% of loop rows have non-memory evidence or mixed evidence.
- At least 80% of loop rows have known urgency and recommended closure window.
- At least 70% of Grade A/B rows have usable review/contact/action/closure or censoring information.
- All accidental identifiers are removed before analysis.
- Exported data can compute days_to_review, days_to_closure, censored_status, days_overdue, risk_weighted_delay, and stall_stage for usable rows.

16. Appendices

Appendix A - Copy-Paste Recruitment Message

Dear Doctor / Clinic Team,

I am conducting a short research pilot on outpatient care-loop progression in Indian clinical settings. The study examines how abnormal diagnostic results and referral/follow-up tasks move from review to patient contact, action, escalation, and closure.

The survey takes about 13-19 minutes. It asks you to audit 6 recent care loops from the last 30-90 days using records where possible. Please do not enter patient names, phone numbers, addresses, Aadhaar numbers, medical record numbers, screenshots, prescriptions, lab-report photos, or other identifying information.

The purpose is research and dataset development, not doctor performance evaluation or commercial marketing.

Survey link: [insert link]

QR code: [insert QR]

Thank you for helping measure where outpatient care progression succeeds or stalls.

Appendix B - Example Completed Loop Row

Field	Example value
loop_id	L014
loop_family	Abnormal diagnostic result
loop_type	Abnormal HbA1c
urgency	Semi-urgent
recommended_window_days	7
review_day_offset	2
contact_day_offset	5
action_day_offset	5
closure_day_offset	14
owner_role	Physician + receptionist/coordinator
handoff_count	2
tracking_method	Paper log
resolution_status	Resolved late
evidence_source	Lab log + paper chart
confidence	High
Derived days_overdue	$14 - 7 = 7$
Derived risk_weighted_delay	$7 * 2 = 14$

Appendix C - Data Quality Grading

Grade	Definition	Use
A	Record-backed, high confidence, complete timeline, specific loop, selection method known.	Main analysis and strongest evidence.
B	Record-backed or mixed evidence,	Main analysis with flags.

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	medium/high confidence, partial timeline but usable closure or censoring information.	
C	Memory-based but tied to one specific loop, usable urgency/status/approximate timing.	Sensitivity analysis only.
D	Vague, missing evidence source, no usable timeline, generic usually answers, or possible patient identifiers.	Drop from main analysis.

Appendix D - API Sketch

POST /api/submissions/start

-> returns respondent_id, submission_id

POST /api/respondent-context

body: respondent context fields

-> saves context

POST /api/care-loops/:loopNumber

body: one care-loop object

-> validates and saves loop draft

GET /api/submissions/:id/status

-> returns complete loops, missing fields, validation warnings

POST /api/submissions/:id/submit

-> final validation and locks submission

GET /api/admin/export/respondents.csv

GET /api/admin/export/care_loops.csv

Appendix E - Final Developer Checklist

- The website has no file upload.
- The website has no patient name, phone, MRN, address, or exact-date field.
- The website requires six complete care loops by default.
- Every loop has urgency, recommended window, closure status, evidence source, confidence, and selection method.
- Day offsets are numeric and validated.
- Care-loop rows export cleanly to CSV.
- Researcher can compute core derived variables from exported data.
- Mobile UX is usable in under 20 minutes.
- Landing and consent copy clearly state research-only, de-identified use.

End of specification.